Marketing Analytics — Group 7 Project

Milestone 1: Problem Definition

Free-to-Paid Conversion Optimization

for a Language Writing & Content Creation Platform

A Marketing Analytics Pipeline Using Behavioral Segmentation and A/B Testing

The Problem

Most free writers on the platform never upgrade to a paid plan not because the product lacks value, but because every user receives the same generic "Upgrade now" prompt, at the wrong moment, regardless of how deeply they actually use the Armenian writing tools.

The Solution

We track how writers behave inside the platform, group them by writing patterns and engagement, and send each group a different, more relevant upgrade message. We then test whether targeted messages outperform the current generic one using a Behavioral Segmentation + A/B Testing pipeline.

1. Problem Area: A/B Testing & Customer Conversion Optimization

Our chosen area of interest lies at the intersection of A/B Testing and Customer Retention. Specifically, we focus on the challenge of free-to-paid conversion within a subscription-based SaaS product: an AI writing and language creation platform built natively for the Armenian language.

This domain is directly relevant to Marketing Analytics because it involves designing controlled experiments, measuring behavioral responses to marketing stimuli, and translating statistical findings into real business decisions. A/B testing is one of the most powerful tools a marketing team can deploy, yet its effectiveness depends entirely on whether the right message reaches the right user at the right moment.

1.1 The Real Reasons Free Writers Don't Convert

- **They all get the same message.** The platform sends every free user the same pop-up: "Upgrade to Pro!" whether the writer just signed up ten minutes ago or has been drafting poetry every day for three months. This is like a bookshop shouting the same sales pitch at every visitor, whether they are browsing for the first time or are a returning author looking for a very specific resource.
- **The message arrives at the wrong time.** The nudge to upgrade often appears randomly — not when the writer is actively frustrated by a limitation. If a poet is in the middle of crafting a poem and hits the paywall on the advanced rhyme-suggestion engine, that is the exact moment they are most likely to upgrade. Missing that moment can be a significant lost opportunity.
- **The message is not relevant to the user's depth of engagement.** A writer who opens the app once a month to caption an Instagram post in Armenian does not care about unlimited manuscript exports or classical synonym layers. But a novelist who writes daily, queries the

thesaurus dozens of times per session, and has hit the vocabulary depth limit repeatedly this week cares deeply. Sending the same pitch to both users is wasteful and erodes trust.

- **There is no follow-up logic.** If a writer sees the upgrade banner and ignores it, nothing changes. They receive the same banner again next session. There is no smarter second attempt — a limited-time discount, a different angle highlighting a specific Armenian literary feature, or a personalized showcase of what they are missing.
- **Dormant writers are completely ignored.** Some users signed up, explored the platform briefly, then stopped returning. These users need an entirely different message — a re-engagement offer that reminds them of their saved drafts and the richness of the Armenian lexical engine — not an immediate payment request.

In short:

The platform is treating all writers as a single group. They are not. A poet working in classical Armenian meters, a blogger writing quick modern opinion pieces, and a novelist building a 300-page manuscript have completely different needs, usage patterns, and reasons to upgrade. A one-size-fits-all approach produces a low conversion rate and wastes the notification budget on messages that do not resonate.

2. Preliminary Research: Trends, Challenges & Opportunities

Before defining the precise problem, we conducted preliminary research across industry benchmarks, SaaS conversion literature, and behavioral marketing studies. Key findings:

- Free-to-paid conversion rates in SaaS products typically range from 2% to 5% on average. Niche-language tools face an even narrower addressable market, meaning every missed conversion carries disproportionate revenue impact.
- Research consistently shows that generic upgrade prompts significantly underperform compared to messages personalized to a user's engagement level and behavior pattern. Personalization can lift conversion rates by 10–15%.
- Behavioral segmentation—grouping users by how they actually use a product rather than demographic proxies—has emerged as a leading technique in retention and conversion marketing. It allows teams to craft messaging that feels relevant rather than intrusive.
- A/B testing is the industry-standard method for validating whether a new messaging approach genuinely outperforms the baseline. Without a controlled experiment, it is impossible to separate the effect of the message from seasonal trends or platform changes.
- For niche-language platforms, the opportunity is particularly acute: Armenian speakers globally number approximately 6–7 million. The digitally active writer and content creator segment is carefully defined, making precision marketing essential rather than optional.

Key Insight from Research

Most niche-language SaaS products lose conversion revenue not because the product lacks value, but because the upgrade communication strategy treats all users identically. The opportunity is not in acquiring new users, it is in converting engaged free users who are already deriving value from the platform.

3. Specific Problem Statement

Problem Statement

Free users of an AI content creation platform are not converting to paid plans at an acceptable rate. The root cause is not product quality but marketing communication: every free user receives the same generic “Upgrade Now” prompt, regardless of their actual engagement depth, usage frequency, or proximity to a paywall moment. This one-size-fits-all approach results in low conversion, notification fatigue, and missed revenue from users who are already engaged but never receive a compelling, contextually relevant reason to subscribe.

This is a specific, measurable problem with a clear scope. We can quantify it through conversion rate data, paywall encounter logs, and session behavior. It is addressable through marketing analytics without requiring product changes. And it has a defined success condition: targeted messaging, validated by a controlled experiment, must statistically outperform the current generic baseline.

The problem breaks into three measurable sub-components:

- Which behavioral patterns distinguish writers who are close to upgrading from those who are not?
- Does a message tailored to a writer’s behavior convert significantly better than a generic prompt?
- Which specific behavioral signals—paywall encounters, session frequency, thesaurus query depth—are the strongest predictors of conversion?

4. Proposed Solution & Methodology

Our solution is a two-stage analytical pipeline: first, behavioral segmentation to group writers by usage patterns; second, a controlled A/B test to determine which upgrade message works best for each group.

4.1 Data Collection

All data is generated organically by the platform during normal user activity. No surveys or external data purchases are required. The following data will be extracted from the platform’s PostgreSQL event database:

Data Type	Specific Fields	Purpose
Session Event Logs	Session timestamps, duration, days since last login	Frequency & recency of engagement
Tool Usage Logs	Thesaurus queries, rhyme-tool uses, synonym depth accessed	Depth of platform engagement
Paywall Encounter Logs	Paywall hits per session, feature blocked, timestamp	Proximity to upgrade moment

Conversion Outcomes	Whether a user subscribed, when, and to which plan	A/B test outcome measurement
Notification Engagement	Upgrade prompt shown, opened, dismissed	Message effectiveness tracking

4.2 Analytical Techniques

The pipeline applies two complementary analytical methods:

K-Means Clustering (Behavioral Segmentation)

We apply K-Means clustering to the per-writer behavioral feature table to automatically discover groups of writers with similar usage patterns. The algorithm does not require us to pre-define what the groups are; it finds them from the data. We expect to find 3–4 stable segments:

Segment	Who They Are	What They Probably Need
Power Writers	Write daily, hit vocabulary/synonym limits often, export or publish frequently	A direct upgrade offers they are already deeply engaged and nearly ready to pay
Growing Creators	Write a few times a week, occasionally hit limits on rhyme suggestions or thesaurus depth	A gentle nudge showing what premium features they are missing
Casual Writers	Write occasionally — personal notes, social captions — rarely hit limits	Not a sales pitch, but a feature discovery message showing advanced Armenian tools
Dormant Writers	Signed up, wrote once or twice, then stopped returning	A re-engagement message — remind them of their drafts and the depth of Armenian vocabulary available

What does clustering do?

It examines each writer's behavioral metrics and automatically groups similar writers together. Writers who use the platform in similar ways—same frequency, same tool depth, same paywall encounter rates—are placed in the same cluster. We do not pre-define what the groups are; the algorithm discovers them from the data.

A/B Testing (Controlled Experiment)

Within each segment, we randomly split users into a 50/50 Control and Treatment group. The Control group receives the current generic upgrade prompt. The Treatment group receives a message crafted specifically for their behavioral profile. The experiment runs for 14 days. Statistical validation uses two techniques:

Control Group (50%)	Treatment Group (50%)
They receive the current generic upgrade message. This is our baseline — the benchmark we are trying to beat.	They receive a message crafted specifically for their segment. For example, a Power Writer might see: "You've queried our Armenian thesaurus 40 times this month — unlock the full classical vocabulary layer for AMD 2,900/month." A Dormant Writer might receive a welcome-back email showcasing their saved drafts and a 20% discount.

After 14 days, we examine the data. For each segment, we compare:

- How many writers in the control group have been upgraded to a paid plan?
- How many writers in the treatment group upgraded to a paid plan?

We apply a Chi-Square Test to verify whether the observed difference is statistically real or could have occurred by chance. If the test confirms statistical significance ($p < 0.05$), we have evidence that the targeted message genuinely outperforms the generic one.

We also use Logistic Regression to identify which behavioral signals — such as paywall hit frequency, thesaurus query depth, or session length — most strongly predict whether a writer will upgrade. This informs how we refine targeting in future campaigns. – This is nice to have in the results.

4.3 Implementation Plan

The analytical pipeline is built on five containerized components, each with a clearly defined role:

- Database (PostgreSQL) Central store for raw event logs, cleaned feature tables, A/B group assignments, segment labels, and conversion outcomes.
- ETL Pipeline (Python) Extracts raw events, computes per-writer behavioral metrics (session frequency, thesaurus depth, paywall counts, days since last login), and loads a clean feature table ready for modeling.
- Model Engine (Python / scikit-learn) — Runs K-Means segmentation and assigns each writer a cluster label. After the test window closes, runs Chi-Square and Logistic Regression to evaluate results.
- API Layer (FastAPI) Exposes model outputs so the platform can query in real time which segment a given writer belongs to and which message to display.
- Dashboard (Streamlit) Visual front-end showing live segment composition, real-time A/B test conversion tracking, statistical significance indicators, and feature importance rankings.

Once the winning message per segment is identified, it is deployed platform wide. The segmentation model reruns weekly. Writers automatically migrate to a new segment—and receive the most relevant messaging—as their behavior evolves.

5. Expected Outcomes

If the pipeline performs as designed, we anticipate the following benefits for the platform's marketing and product teams:

- Higher conversion rate, particularly in the Power Writer segment where personalized messaging about the depth of the Armenian lexical engine has the strongest appeal.
- Reduced notification fatigue — writers stop seeing irrelevant upgrade prompts, which means they are less likely to disable notifications entirely or develop negative associations with the platform's monetization approach.
- A reusable pipeline that can be adapted to future campaigns: re-engagement drives for dormant writers, referral programs targeting prolific users, feature adoption nudges for specific Armenian language tools.
- Data-backed decisions — instead of guessing which message to send, the team has statistical evidence grounded in actual writer behavior.

6. Evaluation Metrics

The success of the solution will be measured against the following Key Performance Indicators (KPIs):

KPI	Definition & Target
Conversion Rate	Percentage of free users who upgrade during the 14-day test window. Primary success metric. Target: statistically significant lift over control baseline ($p < 0.05$).
Segment Conversion Rate	Conversion rate broken down by behavioral segments. Reveals which segments respond most strongly to targeted messaging.
Notification Engagement	Percentage of users who open or interact with the upgrade prompt vs. dismiss. Measures message relevance before conversion.
Statistical Significance	Chi-Square test result per segment A/B comparison. The result of $p < 0.05$ confirms the observed conversion difference is not due to chance.
Customer Lifetime Value	Estimated revenue a converted subscriber will generate over time. Tracks on whether conversion-optimized users have higher long-term value.
Churn Rate Post-Conversion	Percentage of newly converted users who cancel within 30 days. Ensures we are attracting genuinely interested subscribers.
Segmentation Stability	How consistently writers remain in the same behavioral cluster week-over-week. Validates the segmentation model quality.
ROI of Campaigns	Revenue generated by targeted upgrade messages relative to the cost of developing and deploying them. Demonstrates overall business value.

References:

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Note:

This document is a living draft and will be updated as the project progresses through subsequent milestones.

Last edited by PM — Milestone 1